

Availability Concerns -- Guided Notes ANSWER KEY

Unit tp-6.3 | CompTIA Network+ N10-009 Obj. FC0-U71 6.1 | N10-008 FC0-U71 6.1

For instructor use / distribute after students complete the notes.

Question 1

Availability in security means that systems, services, and data are _____ and _____ to authorized users when needed. An availability failure is called a _____ of Service. Causes include hardware failures, software bugs, natural disasters, and deliberate _____.

Answer: Accessible; operational (functioning); denial; attacks.

Question 2

A Denial of Service (DoS) attack floods a target with _____ traffic or requests until it cannot serve legitimate users. A Distributed DoS (DDoS) attack uses thousands of _____ systems (called a _____) to amplify the attack. The compromised machines are typically infected with _____ allowing the attacker to control them remotely.

Answer: Overwhelming (excessive); compromised; botnet; malware.

Question 3

Availability is measured as _____ uptime. 99.9% uptime ('three nines') allows _____ hours of downtime per year. 99.999% ('five nines') allows only _____ minutes per year. Organizations choose their availability target based on the _____ of downtime (cost per hour of outage).

Answer: Percentage; 8.76 hours; 5.26 minutes; business impact (cost).

Question 4

_____ eliminates single points of failure by having duplicate components so that if one fails, the other takes over. This includes redundant _____ supplies, _____ (multiple drives working together), _____ servers (secondary takes over if primary fails), and multiple _____ links to the internet.

Answer: Redundancy; power; RAID; failover; network (ISP).

Question 5

A _____ (BCP) is an organizational plan to maintain critical functions during and after a disaster. The _____ (DR) plan is a subset that focuses specifically on restoring IT systems. RTO (Recovery Time Objective) defines the maximum _____ the organization can tolerate. RPO (Recovery Point Objective) defines the maximum acceptable _____ loss.

Answer: Business Continuity Plan; disaster recovery; downtime; data.

Question 6

Load balancing distributes requests across multiple servers to prevent any one server from being _____ and to maintain availability during _____ spikes. _____ (Content Delivery Networks) cache content at geographically distributed edge nodes, reducing load on origin servers and improving availability globally.

Answer: Overloaded; traffic; CDNs.

Question 7

RAID (Redundant Array of Independent Disks) protects availability of stored data. RAID _____ mirrors all data to two drives -- if one fails, data is still available. RAID _____ stripes data across three or more drives with parity, tolerating one drive failure. RAID _____ requires only raw speed but provides no fault tolerance.

Answer: 1; 5; 0.

Question 8 -- Real World Application

REAL WORLD: A small clinic runs its patient scheduling system on a single physical server with no backup server and no offsite backups. A ransomware attack encrypts all files on the server. The clinic cannot schedule patients and cannot access existing appointment data. Identify two availability failures that made this scenario possible and describe specific controls for each.

Answer: Failure 1 -- No redundancy/failover: the clinic has a single server with no standby. When it was taken offline (encrypted), there was no failover system to maintain operations. Control: implement a hot standby or cloud-based replica that can take over if the primary server is offline. Even a cold standby (a second server that can be restored from backup within a few hours) reduces downtime from days to hours. Failure 2 -- No offline or offsite backups: ransomware specifically targets and encrypts backup files on attached or network drives. A proper backup strategy uses the 3-2-1 rule: three copies of data, two different media types, one copy offsite or in immutable cloud storage (e.g., AWS S3 with object lock). An air-gapped or immutable backup cannot be encrypted by ransomware. With a clean backup from yesterday, the clinic restores the server and loses at most one day of appointment data -- not everything.